CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

M.Tech. (EC)(C.S.E.) SEM – I , Examination, 2010-11

EC - 704 Digital Signal Processing and Applications

Date: 06/01/2011, Thursday **Time: 10.30 a.m to 01.30 p.m** **Maximum Marks: 70**

Instructions:

1. **Attempt all questions.**
2. **Answer to the two sections must be written in separate answer books.**
3. **Figures to the right indicate *full* marks.**
4. **Make suitable assumptions and draw neat figures wherever required.**

SECTION-I

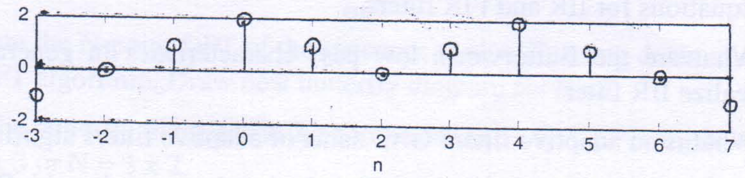
Q-1 a) Define: (i) Discrete time signal (ii) Discrete time system. Write mathematical expressions of discrete time signals and systems. **2**

b) What is the sampling theorem? How to reduce aliasing? **3**

c) How to determine based on ROC (Region of Convergence) that signal (for infinite duration) is causal or anti-causal. Comment on existence of z-transform. **3**

d) Explain: Classification of signals **3**

Q-2 a) Let $X(e^{jw})$ denote the Fourier transform of the signal x[n] shown in below figure and x[n] = 0, else, perform the following calculations without explicitly evaluating $X(e^{jw})$. **8**



(i) Evaluate $X(e^{jw})\big|_{w=0}$. (ii) Evaluate $X(e^{jw})\big|_{w=\pi}$.

(iii) Find phase of $X(e^{jw})$. (iv) Evaluate $\int_{-\pi}^{\pi} X(e^{jw})dw$

b) Determine z-transform and ROC of the following sequences **4**

(i) $x[n] = a^{-n}u[-n]$ (Use time reversal property)

(ii) $x[n] = (0.5)^{n-1}u[n]$

OR

Q-2 a) Determine the frequency response of a linear time-invariant system $y[n] = x[n - n_d]$, where n_d is a fixed integer. $x[n] = e^{jwn}$ as input to this system. **4**

b) For the given sequence $x[n] = (\frac{1}{2})^n u[n]$, determine magnitude $|X(e^{jw})|$ and phase of $X(e^{jw})$. Sketch the magnitude and phase response. 4

c) Find the inverse z-transform of 4

$$X(z) = \frac{z+3}{z^2(z-0.5)}, \text{ where x(n) is causal sequence.}$$

Q-3 a) When the input to a causal LTI system is 8

$$x[n] = -\frac{1}{3}\left(\frac{1}{2}\right)^n u[n] - \frac{4}{3}2^n u[-n-1],$$

the z-transform of the output is

$$Y(z) = \frac{1+z^{-1}}{(1-z^{-1})(1+\frac{1}{2}z^{-1})(1-2z^{-1})}.$$

Determine (i) ROC of Y(z) (ii) Impulse response of the system.

b) A system has an impulse response h[n] = {1,2,3} and output response y[n]={1,1,2,-1,3}. Using Long division method, determine inverse z-transform of X(z), x[n] is causal. 4

OR

Q-3 a) Using residue method, determine the inverse z-transform of 6

$$X(z) = \frac{1}{(z-0.25)(z-0.5)}, \; ROC: |z| > 0.5.$$

b) Prove properties of Z-transform. ***(Any Four)*** 6

SECTION-II

Q-4 a) What is structure? Write Linear Constant Co-efficient Difference Equations for IIR and FIR filters. 2

b) What are the Buttorworth low pass characteristics in general sense to realize IIR filter? 3

c) What is an adaptive filter? Give name of adaptive filters algorithms. 3

d) Short Note: Contribution of digital signal processing in Speech and Image 3

Q-5 a) The sequence $x[n] = \cos(\frac{\pi}{4}n)$, $-\infty < n < \infty$, 4
was obtained by sampling a continuous-time signal

$$x_c(t) = \cos(\Omega_0 t), \; -\infty < t < \infty,$$

at a sampling rate of 1000 samples/second. What are two possible positive values of Ω_0 that could have resulted in the sequence x[n]?

b) Draw the possible cascade and parallel form implementations of the LTI system with system function 4

$$H(z) = \frac{1+2z^{-1}+z^{-2}}{1-0.75z^{-1}+0.125z^{-2}}.$$

c) Draw direct form II and transposed form from the difference equations of the system 4

$$w[n] = a_1 w[n-1] + a_2 w[n-2] + x[n]$$

$$y[n] = b_0 w[n] + b_1 w[n-1] + b_2 w[n-2]$$

OR

Q-5 a) To design a low pass digital FIR filter using Kaiser Window satisfying the specifications given below. 4

Pass band cutoff frequency, f_p= 150 Hz,
Stop band cutoff frequency, f_s = 250 Hz,
Pass band ripple, A_p = 0.1 dB, δ = 0.0057
Stop band ripple, A_s = 40 dB and sampling frequency, F = 1 kHz.

Determine the value of A, β, and M.

b) Design a digital Buttorworth filter that satisfies the following constraint using Bilinear transformation. Assume T = 1sec. 8

$$0.9 \le |H(e^{jw})| \le 1 \qquad 0 \le w \le \tfrac{\pi}{2}$$

$$|H(e^{jw})| \le 0.2 \qquad \tfrac{3\pi}{4} \le w \le \pi$$

Q-6 a) An input sequence x[n] = {1, 1, 2, 2} is applied to a DSP system having an impulse response h[n] = {1, 2, 3, 4}. 8

i) Determine the Circular Convolution of two sequences using DFT and IDFT. Verify with the graphical method.

ii) Determine the Linear Convolution of two sequences using Circular Convolution.

b) What is difference between DTFT and DFT? Write mathematical expressions of DFT and IDFT? State advantages of FFT over DFT. 4

OR

Q-6 a) Compute the N-point DFT of the sequence x[n] = 2^n, using decimation in time FFT algorithm. Draw neat butterfly diagram for N=8. 8

b) Draw the butterfly diagrams using composite-radix DIT FFT, for N = 6. N = 2 x 3 or N = 3 x 2. 4
